Predicting Exoplanet Presence based on Data from Kepler Space Observatory

Introduction

The Kepler Space Observatory (KSO), or Kepler Space Telescope, is a now-deactivated space telescope launched by NASA in 2009 with the purpose of discovering Earth-sized exoplanets outside the solar system. Producing large amounts of data which have been exhaustively studied by NASA and the wider scientific community, it is a perfect candidate for data analysis and prediction. The dataset that will be used for this project is the Kepler Objects of Interest (KOI) Dataset, which can be found on [Kaggle]() and on [NASA's official website]().

The principle of operation for the KSO was to observe stars and wait until a dip in light emission was detected from the star. Then, additional data is collected from the star to predict whether the dip in radiance was caused by the orbit of a planet around it, or whether an exoplanet was not the culprit of the dip. This method of detecting dimming from stars is referred to as the *planetary transit hypothesis* in the field of astronomy.

Data Description, Target, & Task

The dataset contains 10,000 instances which are all exoplanet candidates. Each is an observation of a star which has undergone a dimming characteristic of planetary transits, and thus are worthy of investigation.

From the Kaggle entry, the following columns are noteworthy:

- ***kepoi_name:*** An identifier for the instance.
- ***kepler_name:*** names for confirmed exoplanets.
- ***koi_pdisposition:*** The disposition Kepler data analysis has towards this exoplanet candidate. One of FALSE POSITIVE, NOT DISPOSITIONED, and CANDIDATE.

The dataset contains 50 columns, one of which is entitled ***koi_disposition*** (not to be confused with ***koi_pdisposition***), which is described by NASA as "The disposition in the literature towards this exoplanet candidate. One of CANDIDATE, FALSE POSITIVE, NOT DISPOSITIONED or CONFIRMED."

- An instance marked CANDIDATE is an instance where it has been ruled out that it may be a FALSE POSITIVE
- One marked NOT DISPOSITIONED is an instance for which NASA has not yet completed all testing required to designate a value for this column
- Those marked FALSE POSITIVE or CONFIRMED have undergone all necessary testing by NASA and have been designated respectively

koi_disposition is the *target column.* Since this column may be one of four classes, this is a *classification* task.

NASA has published an extensive data dictionary for the dataset.
Columns are categorized between **Identification, Exoplanet Archive Information, Project Disposition, Transit Properties, Threshold-Crossing Event (TCE) Information, Stellar Parameters, KIC Parameters, and Pixel-Based KOI Vetting Statistics.** Some of these require specialized astronomical knowledge, which will be elaborated on as necessary.

Due to the complexity of the dataset, all 50 fields are described in brief and in lay terms in a separate document entitled Dataset Fields Explanations.

Brief Literature Review & Proposed Solution

A study at Atma Jaya Catholic University of Indonesia entitled ***Exoplanet Classification Through Machine Learning: A Comparative Analysis of Algorithms Using Kepler Data*** used the same dataset and multiple algorithms to compare them, and found multiple algorithms which produced exemplary results, including Decision Tree, XGBoost, and Logistic Regression. The "Ensemble" and "Boosting" models (like Random Forest and LightGBM) performed very well, achieving nearly 99% accuracy. Simple probabilistic models like Naive Bayes performed the worst.

A separate study in the Journal of Student Research entitled ***Using Machine Learning to determine the most important features in exoplanet verification,*** also using the same dataset, selected 15 of the original 50 features of the dataset and used several algorithms to determine which 5 of them were most important for the final prediction of whether an instance represented an exoplanet or not. This study will likely be very helpful for our own.

To solve this classification task, the proposed or preliminary solution will first use a Random Forest Classifier as a baseline model. As shown in the literature, these models are highly effective for this specific dataset. Random Forest allows us to extract feature importance. This will be necessary to determine which astronomical parameters are most critical for the final predictions.

Following the evaluation of the baseline, an XGBoost (Extreme Gradient Boosting) model will be implemented to further optimize accuracy. Due to the complexity of the KOI dataset, XGBoost is a perfect candidate for this project. It handles the imbalanced classes and complex relationships which are common in astronomical datasets very well.